

Adopted Levels, Gammas

$Q(\beta^-) = -16891.17$; $S(n) = 17065.34$; $S(p) = 4663.94$; $Q(\alpha) = -6743.9522$ [2021Wa16](#)

$Q(\beta^-)$: Deduced by the evaluator using mass excesses of -1487.17 for ^{34}K measured by [2024Dr01](#); -18378.28980 for ^{34}Ar from [2021Wa16](#). Value from [2021Wa16](#): $Q(\beta^-) = -17160.200$ (syst).

$S(2n) = 32320.618$, $S(2p) = 6940.708$, $Q(\varepsilon) = 6061.796$, $Q(\varepsilon p) = 918.598$ ([2021Wa16](#)).

Identification discovery ([2012Th10](#)): $^{32}\text{S}(^3\text{He}, n)$ at California Institute of Technology ([1966Mi11](#)).

[1967Ba36](#) in $^{32}\text{S} + ^3\text{He}$ reaction; measured half-life.

$^{34}\text{S}(\pi^+, \pi^-)$: [1993Bi10](#), [1991Bi07](#) ($E = 50$ MeV), [1987Zu03](#) ($E = 292$ MeV), measured σ .

Mass measurement: [2002He23](#) (also [2001He29](#), [2001He37](#)).

Nuclear radius measurement: [2002Oz03](#).

Structure calculations: [2006Or01](#) (levels, $B(E2)$, shell model); [2005Ob01](#) (deformation, levels).

[2007DoZV](#): found 1197γ and 2090γ (with two-step fragmentation reaction at relativistic energies) $^9\text{Be}(^{37}\text{Ca}, X\gamma) E = 197.5$ A MeV reaction).

[2011Le01](#), [2010Le03](#): experimental (by two methods) and theoretical neutron spectroscopic factors and reduction factors for ^{34}Ar g.s. extracted from reaction $p(^{34}\text{Ar}, d)$, $E(^{34}\text{Ar}) = 33$ MeV/nucleon.

 ^{34}Ar LevelsCross Reference (XREF) Flags

A	^{35}Ca εp decay (25.7 ms)	D	$^{32}\text{S}(^3\text{He}, n)$
B	$^1\text{H}(^{34}\text{Ar}, p')$	E	$^{36}\text{Ar}(p, t)$
C	$^3\text{He}(^{32}\text{S}, n\gamma)$, $^{32}\text{S}(^3\text{He}, n\gamma)$		

E(level)	J^π	$T_{1/2}^\dagger$	XREF	Comments
0	0^+	843.8 ms 4	ABCDE	$\% \varepsilon + \% \beta^+ = 100$ $\langle r^2 \rangle^{1/2} = 3.365$ fm 4 (2004An04 evaluation and its 2008 update on webpage: http://cdfc.sinp.msu.ru). $\delta \langle r^2 \rangle (^{38}\text{Ar}, ^{34}\text{Ar}) = -0.251$ fm ² 6 62, first is statistical and the second is systematic uncertainty (1996K104 , 2000Ge20). $T_{1/2}$: from 2006Ia05 , half-life measured and analyzed using parent-daughter (^{34}Ar to ^{34}Cl decay) composite decay and a new fitting procedure, gas-ionization chamber used as detection system. Beam of pure ^{34}Ar ions was produced in $^1\text{H}(^{35}\text{Cl}, 2n)$ reaction. Others: 844.5 ms 34 (1974Ha26 , also 1972Ha58), 0.85 s 10 (1967Ba36), 1.2 s 3 (1966Mi11). $\beta_2(p, p') = 0.272$ (2001Kh17). J^π : $L(p, t) = L(p, p') = 2$. $T_{1/2}$: 2001Ra27 evaluation lists 305 fs 49 which is close to the value given here from 1985Al18 . It seems a somewhat different averaging procedure is used in 2001Ra27 .
2091.1 3	2^+	319 fs 42	ABCDE	$\beta_2(p, p') = 0.272$ (2001Kh17). J^π : $L(p, t) = L(p, p') = 2$. $T_{1/2}$: 2001Ra27 evaluation lists 305 fs 49 which is close to the value given here from 1985Al18 . It seems a somewhat different averaging procedure is used in 2001Ra27 .
3287.7 5	2^+	194 fs 35	A CDE	J^π : $L(p, t) = 2$.
3873 3	0^+	>187 fs	A CDE	J^π : $L(p, t) = 0$.
4050 14			DE	
4127.8 10		<208 fs	CD	
4513.2 8	3^-	201 fs 38	BCDE	$\beta_3(p, p') = 0.393$ (2001Kh17). J^π : $L(p, p') = L(^3\text{He}, n) = 3$.
4631 4			C E	
4865 4			C E	
4967 4	0^+		CDE	J^π : $L(p, t) = 0$.
5255? 4			C	
5307 13	(5^-)		DE	J^π : $L(^3\text{He}, n) = (5)$.
5542 4			C	
5620 30	2^+		D	J^π : $L(^3\text{He}, n) = 2$.
5909 12	0^+		DE	J^π : $L(^3\text{He}, n) = 0$.
6074 11	2^+		E	J^π : $L(p, t) = 2$.

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{34}Ar Levels (continued)

E(level)	J^π	XREF	Comments
6525 9	2^+	DE	J^π : L(p,t)=2.
6794 11		DE	
6990 50		D	
7322 6	2^+ (2^+)	DE	J^π : L(p,t)=2. J^π : L(p,t)=(2).
7499 4		E	
7925 5		E	

[†] From DSAM in $^3\text{He}(^{32}\text{S},n\gamma), ^{32}\text{S}(^3\text{He},n\gamma)$. Most values are from 1985A118.

 $\gamma(^{34}\text{Ar})$

$E_i(\text{level})$	J_i^π	E_γ [†]	I_γ [†]	E_f	J_f^π	Mult. [†]	δ [†]	Comments
2091.1	2^+	2091.1 3	100	0	0^+	E2		B(E2)(W.u.)=6.8 9
3287.7	2^+	1196.6 4	100 5	2091.1	2^+	M1+E2	+0.12 5	B(M1)(W.u.)=0.060 12; B(E2)(W.u.)=2.4 20
		3286 4	9 6	0	0^+	[E2]		B(E2)(W.u.)=0.10 7
3873		585 [#]	<43	3287.7	2^+			
	0^+	1782 3	100	2091.1	2^+	[E2]		B(E2)(W.u.)<21
4127.8		840.1 9	100 6	3287.7	2^+			
		2037	11 6	2091.1	2^+			
	3^-	4128 [#]	<11	0	0^+			
4513.2		1225.5 6	100 11	3287.7	2^+	[E1]		B(E1)(W.u.)=0.0016 4
		2422	11 3	2091.1	2^+	[E1]		B(E1)(W.u.)=2.2×10 ⁻⁵ 8
4631	0^+	2540	100	2091.1	2^+			%I γ >50.
4865		2774	100	2091.1	2^+			%I γ >50.
4967		841 [#]	<20	4127.8				
		2876	100	2091.1	2^+			%I γ >50.
5255?		3164 [#]	100	2091.1	2^+			%I γ >50.
5542		911	54 [‡] 16	4631				
		1029	100 [‡] 16	4513.2	3^-			

[†] From $^3\text{He}(^{32}\text{S},n\gamma), ^{32}\text{S}(^3\text{He},n\gamma)$.

[‡] Tentative value of branching ratio.

[#] Placement of transition in the level scheme is uncertain.

Adopted Levels, Gammas

Legend

Level Scheme

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)